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digestion with the help of symbiotic bacteria. The colon is divided into four regions – an ascending, a transverse, a descending part and a sigmoid colon. The colon is lined by dilations called **haustra** (singular – haustrum) (Figure.5.4). The "S" shaped sigmoid colon (pelvic colon) opens into the rectum. Rectum is concerned with temporary storage of faeces. The rectum open out through the anus. The anus is guarded by two anal sphincter muscles. The anal mucosa is folded into several vertical folds and contains arteries and veins called anal columns. Anal column may get enlarged and causes **piles** or **haemorrhoids**.

5.1.2 Histology of the Gut

The wall of the alimentary canal from oesophagus to rectum consists of four layers (Figure 5.5) namely serosa, muscularis, sub-mucosa and mucosa. The serosa (visceral peritoneum) is

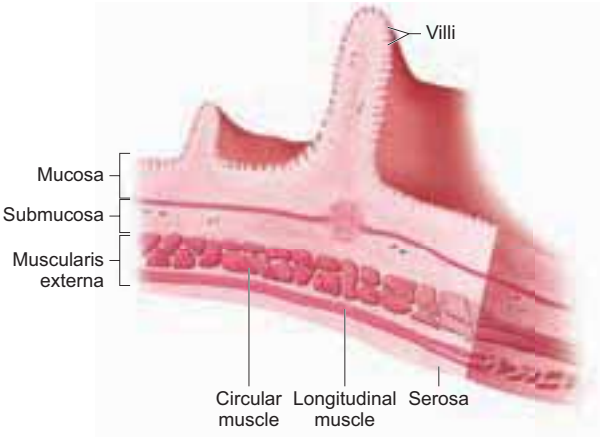


Figure 5.5 The layers of the alimentary canal

the outermost layer and is made up of thin squamous epithelium with some connective tissues. Muscularis is made of smooth circular and longitudinal muscle fibres with a network of nerve cells and parasympathetic nerve fibres which controls peristalsis. The submucosal layer is formed of loose connective tissue containing nerves, blood, lymph vessels and the sympathetic nerve fibres that control the secretions of intestinal juice.

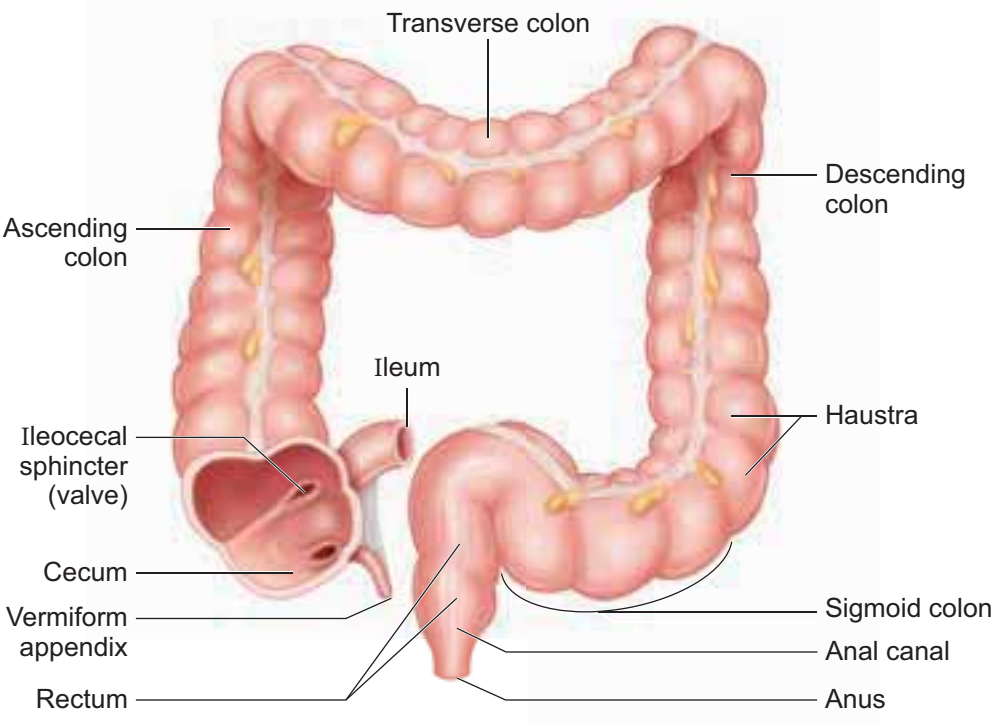


Figure 5.4 Large intestine